

COMP1551 Application Development

Education Centre Desktop Information System

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Submission	Tasks 1 & 2 — SRS and UML Diagrams

Task 1 — System Description

An education centre currently relies on manual, paper-based methods for recording and maintaining information about its personnel and students. To modernise this process, a dedicated desktop application — the Desktop Information System (DIS) — will be introduced to consolidate all record management into a single, easy-to-use platform running on Microsoft Windows.

From an operational standpoint, a staff member opens the application on a Windows PC or laptop and is presented with a numbered menu. Through this menu, new entries can be created for any of three categories: teaching staff, administrative personnel, and students. Each category captures basic contact details that apply universally — the person's full name, phone number, and email address — together with information unique to that category. A teacher's record additionally holds their annual pay and the two modules they deliver. An administrator's record stores pay, whether the contract is full-time or part-time, and the number of hours contracted per week. A student's record lists the three subjects they are enrolled on.

Beyond data entry, the system lets the operator retrieve records in two ways: viewing the entire dataset at once, or narrowing the display to a single category. Either way, records are presented in a clear, numbered format. When details change — a pay rise, a new subject, an updated phone number — the operator picks the relevant entry by its number and edits whichever fields need updating, with the stored values shown on screen as a reference. Records that are no longer needed can be permanently removed, though the system requests confirmation first so that nothing is deleted by mistake.

The application is entirely self-contained: it runs on a standard Windows 10 or 11 machine without requiring a network connection, a database server, or any software beyond the .NET runtime. It is built for everyday use by office staff and requires no technical expertise to operate.

Task 1 — Software Requirements Specification

1. Introduction

1.1

Purpose

This document specifies the requirements for the Desktop Information System (DIS) to be developed for the education centre. It establishes both what the software must do and the conditions under which it must operate, serving as the agreed basis for development, testing, and final acceptance.

1.2

Project

Scope

The DIS covers the creation, retrieval, modification, and removal of records belonging to three personnel categories: teaching staff, administrative staff, and students. It is delivered as a standalone Windows console application. Out of scope are user login mechanisms, persistent on-disk storage, and any form of

network communication.

2. Overall Description

2.1

Product

The DIS is a C# console application. All records are held in a dynamic in-memory collection for the duration of each session and are accessible through a numbered, text-based navigation menu.

2.2

Users

The application is operated by administrative staff responsible for maintaining accurate personnel and student records. No prior programming knowledge or specialist IT training is assumed.

2.3

Operational

Environment

The system must run on Windows 10 or Windows 11 on a PC or laptop with a minimum of 4 GB RAM, a standard keyboard, and a monitor. No internet access or external database is required; the only dependency is a compatible .NET runtime installation.

3. System Features

3.1

Description

Three distinct record types are managed — Teacher, Admin, and Student — each inheriting a common set of attributes (name, telephone, email, role) and extending it with category-specific data fields.

3.2 Functional Requirements

- REQ-01: A new record of any category may be created by the operator at any time.
- REQ-02: The operator may request a listing of every record currently held in the system.
- REQ-03: Records may be filtered and displayed by category (teacher, administrator, or student).
- REQ-04: Any field of any existing record may be updated by the operator.
- REQ-05: Any record may be permanently removed, subject to an explicit confirmation step.
- REQ-06: Following each completed operation, control returns to the main menu automatically.

4. User Interface Requirements

Navigation is provided through a numbered console menu. When editing a record, the current stored value is displayed alongside each prompt so the operator can make informed changes. A yes/no confirmation is mandatory before any record is erased.

5. Platform Requirements

Compatibility with Microsoft Windows 10 and Windows 11 is required. The application must run without network access. The only external dependency permitted is the .NET runtime.

6. Quality Attributes

6.1

Performance

Every menu interaction and data operation must complete within one second across datasets of up to one thousand records.

6.2

Security

No authentication mechanism is implemented. Physical access to the workstation is considered sufficient access control for the centre's operational context.

6.3

Safety

A mandatory confirmation prompt must precede any destructive operation to eliminate the risk of unintended data loss.

Task 2 — Class Diagram

```
«abstract» Person – name : string – telephone : string – email : string – role : string + Display() :  
void + Edit() : void {abstract}  
Teacher – salary : double – subject1 : string – subject2 : string +  
Display() : void + Edit() : void + Salary, Subject1, Subject2  
Admin – salary : double – isFullTime :  
bool – workingHours : double + Display() : void + Edit() : void + Salary, IsFullTime,  
WorkingHours  
Student – subject1 : string – subject2 : string – subject3 : string + Display() : void +  
Edit() : void + Subject1, Subject2, Subject3
```

Figure 1 — Class structure showing Person as the abstract base and its three concrete subclasses

Task 2 — Use Case Diagram

DIS — Desktop Information System
Create New Record
Choose Personnel Category
Browse All Records
Browse by Category
Modify Existing Record
Remove a Record
Verify Removal Action
Operator «include» «include»

Figure 2 — Use case diagram illustrating operator interactions with the DIS boundary